

Abraham Mekonnen

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EDUCATION

Eastern Mennonite University Harrisonburg, VA
Bachelor of Science in Computer Science, Minor in Business; Dean's List - U.S. Citizen Aug. 2022 – May 2026
Coursework: Data Structures & Algorithms, Databases & Information Management, Architecture & OS, Web Applications, Networking & Data Communication, Discrete Mathematics, Inferential Statistics

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML/CSS
Frameworks: Angular, Spring Boot, React, Django, Flask, GraphQL
Tools: Git, Docker, Kubernetes, Kafka, MongoDB, RESTful APIs, Prometheus, Grafana, Apache POI, BeautifulSoup, Firebase

EXPERIENCE

Incoming Software Engineering Intern (Return Offer) Feb. 2026 – Jun. 2026
Morgan Stanley New York, NY

Software Engineering Intern Jun. 2025 – Aug. 2025
Morgan Stanley New York, NY

- Contributed to full-stack development of RMD Bookview application using **Angular**, **TypeScript**, and **Spring Boot**, implementing a responsive data grid and export functionality that reduced financial advisor lookup time by 65% across a platform serving 5,000+ users managing retirement assets.
- Developed and integrated Excel export capability using **GraphQL** and **Apache POI** as part of a cross-functional team, enabling the extraction of filtered retirement account data that improved compliance reporting efficiency by 30%.
- Led production deployment of **Grafana** monitoring infrastructure using **PromQL/Prometheus** metrics to track **SQL Server** and **DB2** performance, memory usage, and CPU trends for Retirement Domain APIs on **OpenShift**, reducing system downtime by 85% and enabling proactive scaling before 4GB memory limits.

IT Helpdesk Technician Jul. 2024 – May 2025
Eastern Mennonite University Harrisonburg, VA

- Secured over 1,000 student, faculty, and alumni accounts using **Azure** to monitor and prevent breaches.
- Built over 10 macro scripts to automate spreadsheet data entry, resulting in a 50% reduction in data entry time.

Python and Java Tutor Oct. 2023 – Dec. 2024
Eastern Mennonite University Harrisonburg, VA

- Supported 20+ students in understanding Python concepts through small-group study sessions, earning a 94% satisfaction rate.
- Tailored teaching methods to individual needs, resulting in a 25% improvement in student comprehension of Python concepts.

Computer Science Teaching Intern May 2023 – Aug. 2023
Church World Service (CWS) Harrisonburg, VA

- Supervised the Two-Ville Programming Camp for 40+ high school students, implementing coding challenges and peer programming sessions that boosted student confidence and earned a 90% satisfaction rate.
- Taught programming fundamentals to 25+ adult refugees and configured Xfinity Essentials for their homes.

PROJECTS AND AWARDS

Figma2Deploy | *Python, Kafka, Docker, Kubernetes, Next.js, Figma API, Gemini API* | *Ongoing*

- Architecting a distributed multi-agent AI system that converts Figma designs into deployed Next.js applications, coordinating five specialized agents through Kafka message queues for asynchronous processing.
- Developing event-driven microservices pipeline with Docker containerization and Kubernetes orchestration, enabling autonomous design parsing, code generation, styling, validation, and deployment to Vercel.

FootPrint Mayhem | *React, Firebase, Express, Tailwind CSS, Git*

- Won "Best Use of Defang" prize at HackHarvard 2024's 36-hour hackathon, developing a gamified carbon footprint tracking application that promotes sustainability through streak systems, leaderboards, and point-based rewards.
- Engineered user authentication with Firebase, implemented dynamic scoring mechanics for eco-friendly actions, and designed responsive UI components using Tailwind CSS for seamless cross-platform experience.

MajorToCareer | *Python, TypeScript, BeautifulSoup, OpenAI API, React*

- Built an application to help college freshmen discover career paths by displaying jobs available for given college majors.
- Designed and implemented an interactive UI with continuous scrolling and visual features using React.
- Web-scraped 5,000+ live jobs using BeautifulSoup and clustered them to respective majors with OpenAI API.

LEADERSHIP

President Jan. 2023 – Present
Computer Science Career Club (CSCC) Eastern Mennonite University

- Led weekly skill-building sessions for 15+ CS/CE majors on resume writing, LinkedIn optimization, and LeetCode preparation, resulting in an 8% increase in club membership.

Treasurer and Ambassador Sep. 2023 – Present
International Student Organization (ISO) Eastern Mennonite University

- Planned weekly cultural exchange events for 50+ members to bridge connections between international and domestic students.

NOTABLE FELLOWSHIPS & ORGANIZATIONS

Fellowships: Grow with Google Scholar, FirstGenU Scholar, Discover Explore Fellow, AT&T Tech Academy Fellow
Organizations: ColorStack, National Society of Black Engineers (NSBE), AfroTech